

The sharp Rider gap for split compact metabelian groups

Candidate partial solution for arXiv:1110.6683
agent_lane_02 (GPT5.6)

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Status: candidate partial result, likely valid. The theorem below proves the conjecturally sharp universal constant for every compact semidirect product of two abelian groups. It does not settle the source question for arbitrary compact groups.

1 Source question

The source is M. Alaghmandan, Y. Choi and E. Samei, *ZL-amenability and characters for the restricted direct products of finite groups*, arXiv:1110.6683v4 (2013) [1]. Theorem 2.3 recalls Rider’s estimate for a finite linear combination ψ of irreducible characters on a compact group: if $\psi * \psi = \psi$ and $\|\psi\|_1 > 1$, then $\|\psi\|_1 \geq 301/300$. Immediately afterwards, Remark 2.4 asks:

“We also note that $301/300$ is not sharp; it appears to be an open problem to determine the best constants here.”

The passage occurs on physical PDF page 4 (source Section 2.2):

ZL-amenability and characters for the restricted direct products of finite groups

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THEOREM 2.3 (Rider; see [14] Lemma 5.2). *Let G be a compact group, λ a Haar measure on it, and ψ a finite linear combination of irreducible group characters on G . Suppose that $\psi * \psi = \psi$ and that*

$$\int_G |\psi(x)| d\lambda(x) > 1.$$

Then

$$\int_G |\psi(x)| d\lambda(x) \geq \frac{301}{300}.$$

REMARK 2.4. Rider’s result is stated for the case where $\lambda(G) = 1$. However, a simple rescaling argument shows that this is equivalent to the formulation we have given. We also note that $301/300$ is not sharp; it appears to be an open problem to determine the best constants here.

The following lemma will be used later, in several places. We would like to thank the

We use normalized Haar measures. This is equivalent to the source formulation by the rescaling noted in its Remark 2.4. For an abelian compact group, Saeki proved the sharp alternative

$$\|f\|_1 = 1 \quad \text{or} \quad \|f\|_1 \geq c_S := \frac{1 + \sqrt{2}}{2}$$

for nonzero idempotent measures [2]. The question is whether the same constant works in the noncommutative setting.

2 Proof intuition

Let $G = A \rtimes H$, where both A and H are compact abelian. The irreducible representations of G are the familiar little-group representations. An H -orbit $\mathcal{O} \subset \widehat{A}$, together with a character of its stabilizer, labels one irreducible representation. The key observation is that its minimal central idempotent $d_\pi \chi_\pi$ is, point by point on the underlying space $A \times H$, an ordinary sum of distinct characters of the *abelian* direct product $A \times H$. The associated character rectangles are disjoint and partition $\widehat{A} \times \widehat{H}$.

Consequently, every finite-spectrum central idempotent on the nonabelian semidirect product has exactly the same values, and hence exactly the same L^1 norm, as an idempotent trigonometric polynomial on the abelian direct product. Saeki's sharp abelian theorem applies without loss in the constant. This transference is special to the split abelian-by-abelian structure; the irreducible dual of a general compact group does not admit the required disjoint character rectangles.

3 The transference identity

Let A, H be compact abelian groups, let $\alpha : H \rightarrow \text{Aut}(A)$ be a continuous action, and put $G = A \rtimes_\alpha H$. We identify the underlying probability spaces of G and the abelian direct product $B = A \times H$; in both cases normalized Haar measure is $m_A \times m_H$.

For $\gamma \in \widehat{A}$, write

$$\mathcal{O}_\gamma = H \cdot \gamma, \quad H_\gamma = \{h \in H : h \cdot \gamma = \gamma\}, \quad d_\gamma = [H : H_\gamma].$$

The orbit is finite: it is a compact subset of the discrete group $\widehat{A}$. For $\eta \in \widehat{H_\gamma}$, let

$$\pi_{\gamma, \eta} = \text{Ind}_{A \rtimes H_\gamma}^{A \rtimes H}(\tilde{\gamma} \otimes \eta), \quad \tilde{\gamma}(a, h) = \gamma(a).$$

This is well-defined because h fixes γ on H_γ .

Lemma 1 (Rectangle lemma). *Choose one γ from each H -orbit in $\widehat{A}$. The representations $\pi_{\gamma, \eta}$, with $\eta \in \widehat{H_\gamma}$, form $\text{Irr}(G)$. For*

$$X_{\gamma, \eta} := \mathcal{O}_\gamma \times \{\xi \in \widehat{H} : \xi|_{H_\gamma} = \eta\} \subseteq \widehat{A} \times \widehat{H},$$

the sets $X_{\gamma, \eta}$ are finite, pairwise disjoint, and partition $\widehat{A} \times \widehat{H}$. Moreover, for every $(a, h) \in A \times H$,

$$d_{\pi_{\gamma, \eta}} \chi_{\pi_{\gamma, \eta}}(a, h) = \sum_{(\delta, \xi) \in X_{\gamma, \eta}} \delta(a) \xi(h). \quad (1)$$

Proof. We first recall the short little-group argument. If π is irreducible, its restriction to A splits into finitely many weight spaces. Normality of A makes H permute the weights, and irreducibility makes this action transitive. For a weight γ , the abelian stabilizer H_γ acts diagonally on its weight space. The G -span of any common eigenline is invariant; irreducibility therefore forces the weight space to be one-dimensional and shows that π is precisely one of the displayed induced representations. This also proves completeness and the usual uniqueness up to the H -orbit of γ .

The induced character formula gives, with $d = d_\gamma$,

$$\chi_{\pi_{\gamma, \eta}}(a, h) = \begin{cases} \eta(h) \sum_{\delta \in \mathcal{O}_\gamma} \delta(a), & h \in H_\gamma, \\ 0, & h \notin H_\gamma. \end{cases} \quad (2)$$

Indeed, $A \rtimes H_\gamma$ has index d , and conjugation by the H -coset representatives produces exactly the weights in $\mathcal{O}_\gamma$.

Restriction $\widehat{H} \rightarrow \widehat{H}_\gamma$ is onto by Pontryagin duality. The fiber over η is a coset of the annihilator $H_\gamma^\perp$, which has d elements. Finite-character orthogonality therefore gives

$$\sum_{\substack{\xi \in \widehat{H} \\ \xi|_{H_\gamma} = \eta}} \xi(h) = \begin{cases} d\eta(h), & h \in H_\gamma, \\ 0, & h \notin H_\gamma. \end{cases}$$

Multiplying this identity by the orbit sum in (2) proves (1), since $d_{\pi_\gamma, \eta} = d$.

For a fixed orbit, the restriction fibers partition $\widehat{H}$; distinct H -orbits in $\widehat{A}$ are disjoint. Hence the rectangles are pairwise disjoint and partition $\widehat{A} \times \widehat{H}$. $\square$

Theorem 2 (Sharp Rider gap for split compact metabelian groups). *Let $G = A \rtimes H$, where A and H are compact abelian groups. Let ψ be a finite linear combination of irreducible characters of G satisfying $\psi * \psi = \psi$ with respect to normalized Haar measure. Then*

$$\|\psi\|_1 \in \{0, 1\} \quad \text{or} \quad \|\psi\|_1 \geq \frac{1 + \sqrt{2}}{2}.$$

The positive gap constant is sharp even among nonabelian finite groups of this form.

Proof. Peter–Weyl orthogonality says that a finite-spectrum central convolution idempotent has the unique form

$$\psi = \sum_{\pi \in S} d_\pi \chi_\pi$$

for a finite set $S \subseteq \text{Irr}(G)$. Let

$$E = \bigsqcup_{\pi_\gamma, \eta \in S} X_{\gamma, \eta} \subseteq \widehat{B}, \quad F(a, h) = \sum_{(\delta, \xi) \in E} \delta(a) \xi(h).$$

The union is disjoint by Lemma 1. Thus F is an idempotent trigonometric polynomial on the compact abelian group $B = A \times H$. Equation (1), summed over S , gives the *pointwise* identity

$$\psi(a, h) = F(a, h) \quad ((a, h) \in A \times H),$$

where the left side uses the semidirect multiplication and the right side the direct product multiplication. The normalized Haar measures coincide on the underlying space, so $\|\psi\|_1 = \|F\|_1$. Saeki’s sharp abelian norm theorem [2] now yields the stated alternative.

For sharpness take $G = D_{16} = C_8 \rtimes C_2$, with inversion action. Let π be the two-dimensional irreducible representation associated to the orbit $\{1, -1\} \subset \widehat{C_8}$. For $\psi = 2\chi_\pi$, one has

$$\psi(r^k) = 4 \cos(\pi k/4), \quad \psi(sr^k) = 0,$$

and hence

$$\|\psi\|_1 = \frac{1}{16} \sum_{k=0}^7 |4 \cos(\pi k/4)| = \frac{1 + \sqrt{2}}{2}.$$

$\square$

4 Verification and limitations

The reusable script `code/verifier.py` checks the rectangle identity and partition property for 3,160 little-group blocks in cyclic semidirect products $C_n \rtimes C_m$ with $2 \leq n \leq 14$, $2 \leq m \leq 6$, over every admissible cyclic action in that range. It also verifies the exact D_{16} equality and exhaustively checks all central idempotents of D_{2n} for $3 \leq n \leq 30$. These computations are sanity checks, not a substitute for Lemma 1. Details and the exact command appear in `VERIFIER.REPORT.md`.

The original question ranges over all compact groups. The proof here requires a *split* extension with both normal factor and complement abelian. For a general compact group, irreducible representations do not yield a disjoint partition of an abelian dual into character rectangles, so the transference step is unavailable.

Novelty check. A bounded search on 11 August 2026 used the exact source phrase, “Rider” with “best constant”, “central idempotent” with “semidirect product”, “metabelian”, and “abelian normal subgroup”, plus forward citations visible from the source. The closest structural paper found was Hauenschild’s 1979 characterization of central idempotent measures for groups with an abelian normal subgroup of finite index [3]; its theorem is phrased through hypercoset rings and the inspected six-page text does not state this norm-preserving direct-product transference or a sharp gap. A 2023 thesis still records the sharp finite-group bound as open and reports computations supporting `cs` [4]. Thus this packet claims a new partial theorem with *moderate*, not definitive, novelty confidence; it may also be a previously unrecorded corollary of older structure theory.

Human-review recommendation. Check first the compact little-group classification and the restriction-fiber orthogonality in Lemma 1. Then check that the rectangles remain disjoint when several irreducibles are summed. Once those points are accepted, the reduction to Saeki is exact and the sharpness computation is elementary.

References

- [1] M. Alaghmandan, Y. Choi and E. Samei, *ZL-amenability and characters for the restricted direct products of finite groups*, arXiv:1110.6683v4, 2013.
- [2] S. Saeki, *On norms of idempotent measures*, Proc. Amer. Math. Soc. **19** (1968), 600–602, doi:10.1090/S0002-9939-1968-0225102-6.
- [3] W. Hauenschild, *Zentrale idempotente Maße auf lokalkompakten Gruppen mit Abelschem Normalteiler von endlichem Index*, Manuscripta Math. **28** (1979), 405–410.
- [4] J. Sawatzky, *Mind the GAP: Amenability*, Master’s thesis, University of Waterloo, 2023, Chapter 6, Conjecture 6.2.3.